



NATURAL CLIMATE VARIABILITY

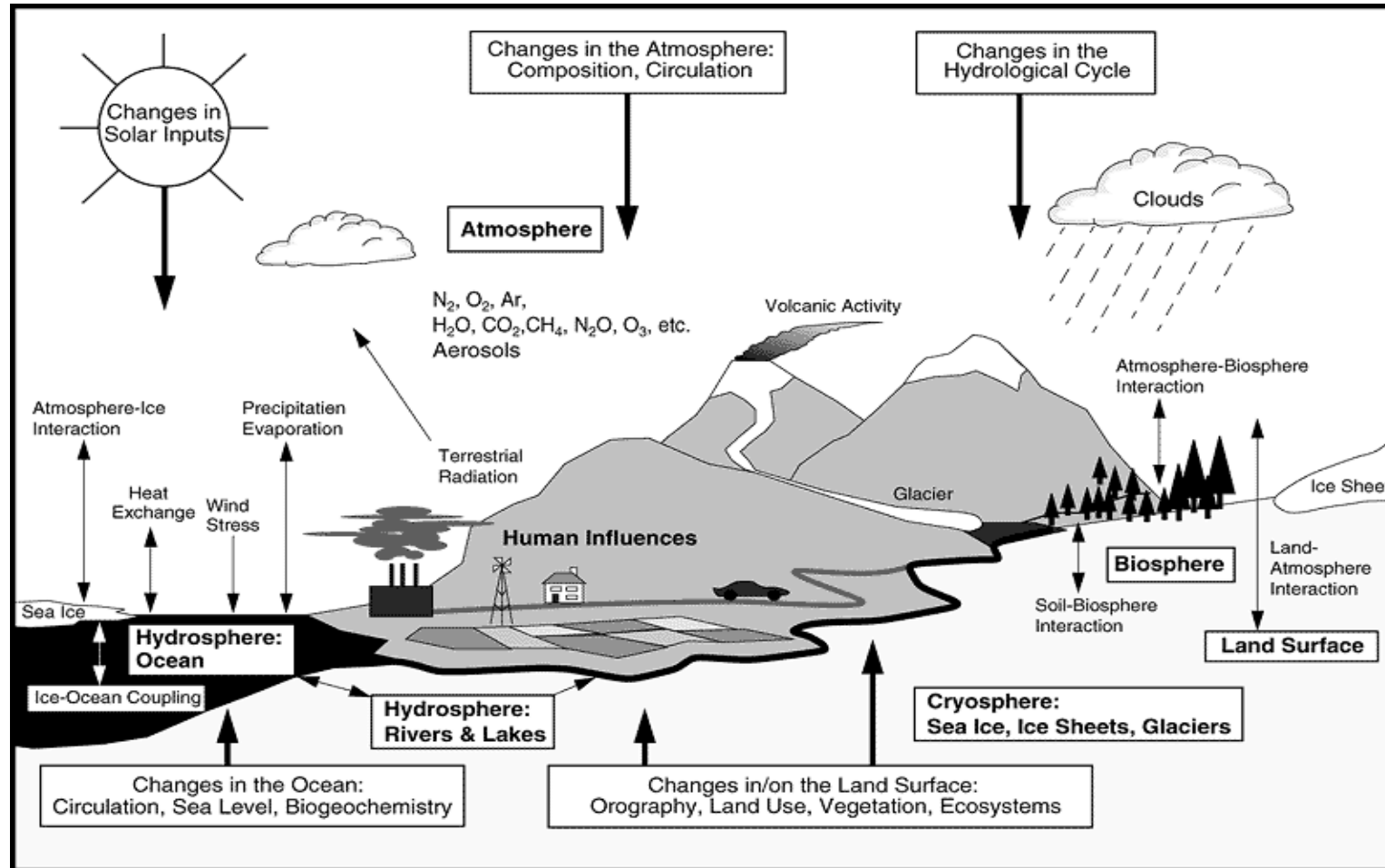
Dr Phikolomzi Matikinca

Department of Environmental and Geographical Science (EGS)
University of Cape Town (UCT)
Office: 4.08

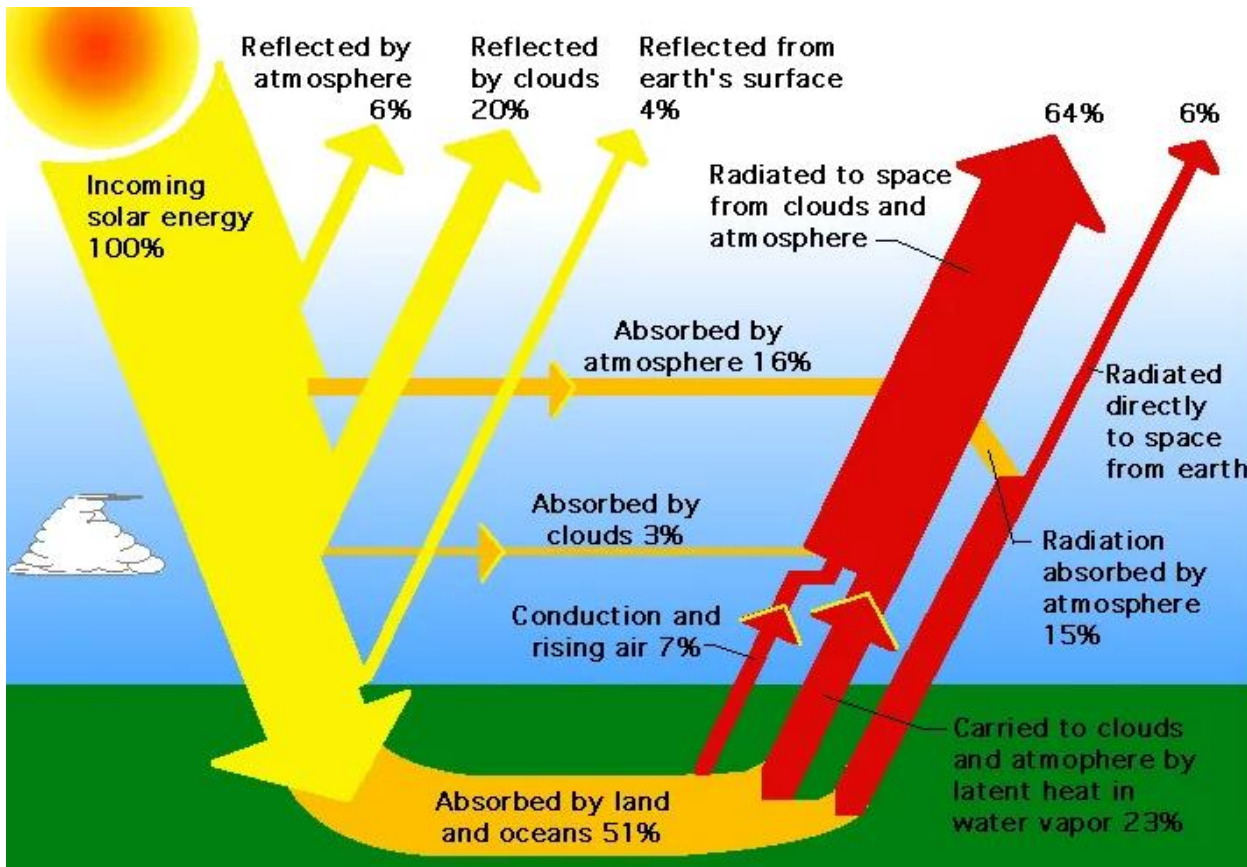
Email: phikolomzi.matikinca@uct.ac.za

EARTH'S CLIMATE SYSTEM

- Complex, interactive network driven by solar energy
- Its stability depends on **Earth's Energy Budget**
 - Balance between incoming shortwave energy from the Sun and outgoing longwave energy from Earth



EARTH'S ENERGY BUDGET



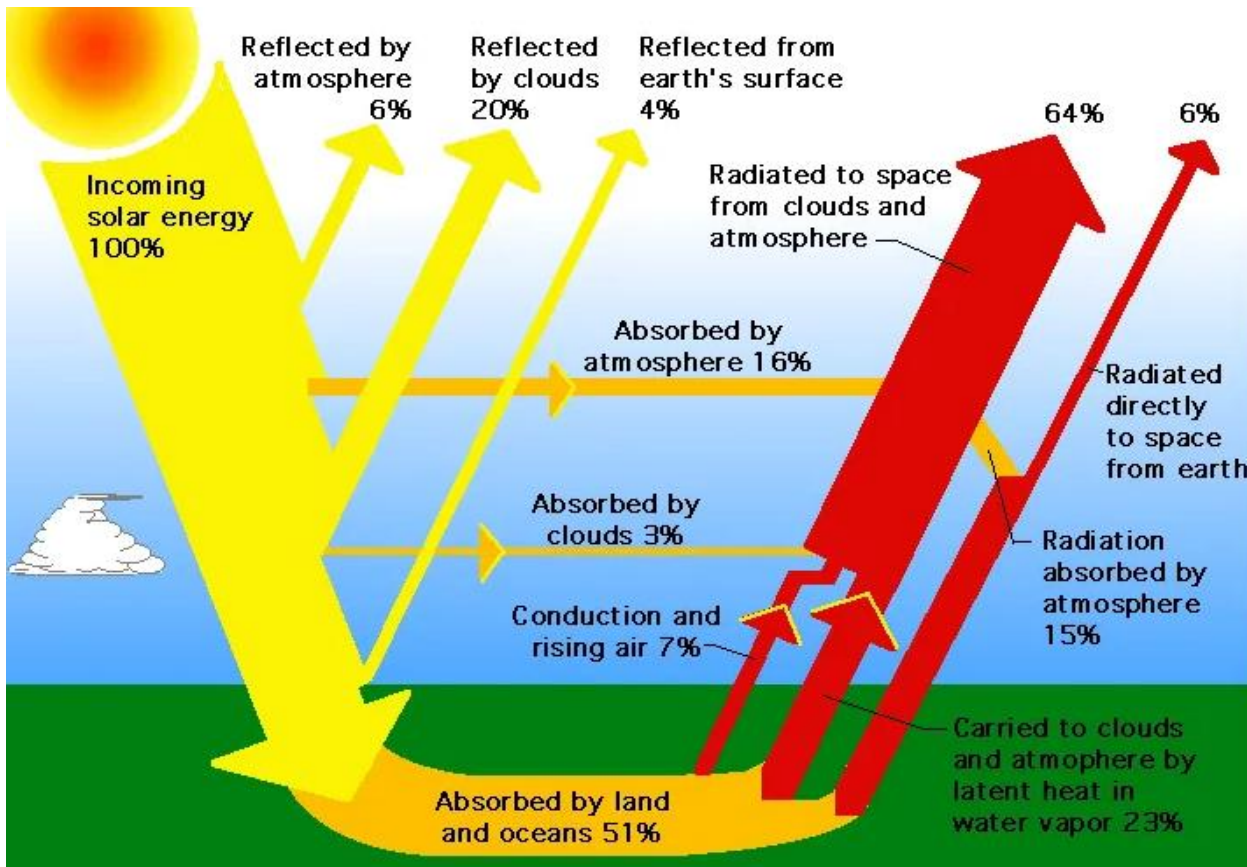
Energy In

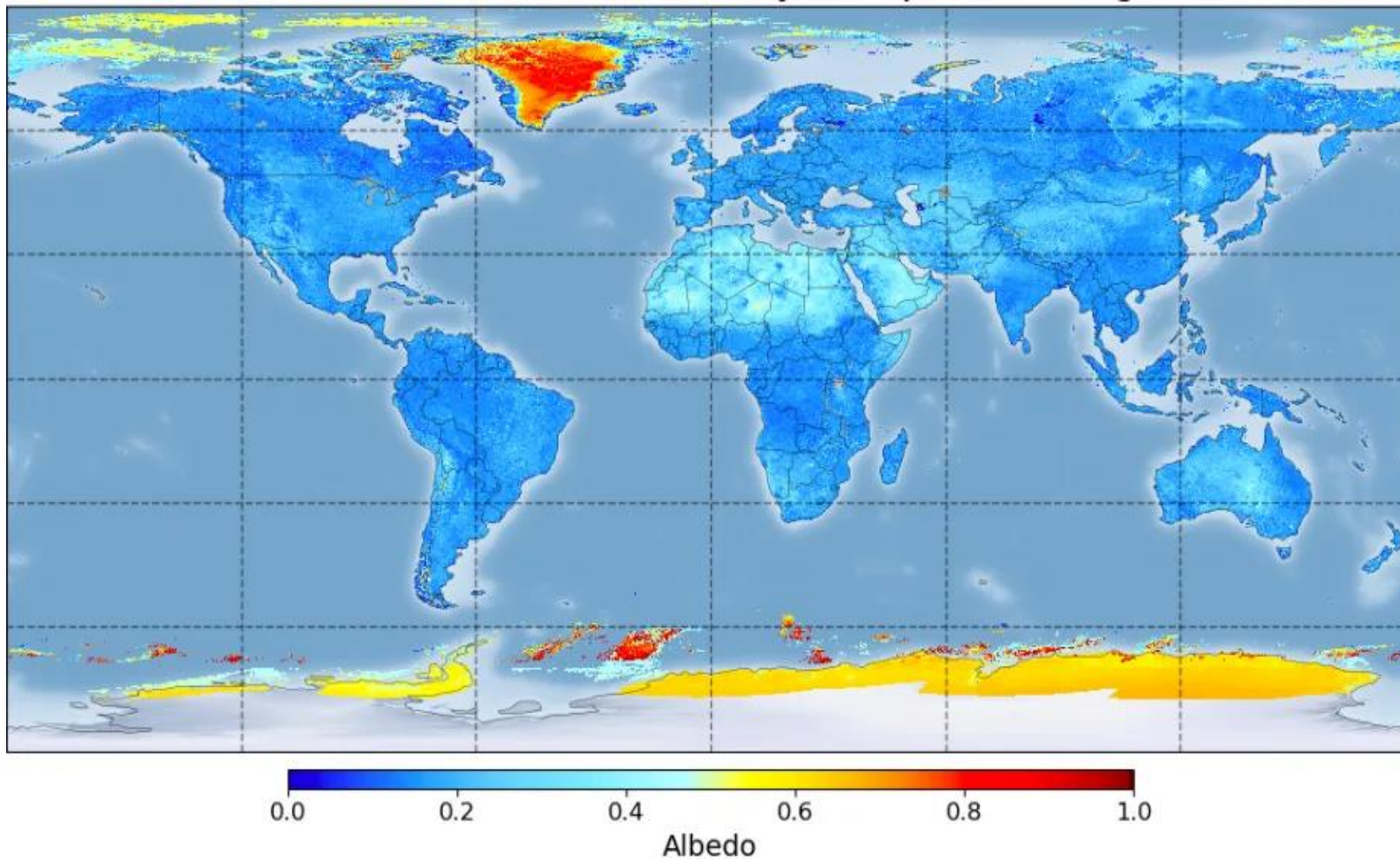
- **Solar Radiation:** The Sun provides Earth with most of its energy in the form of visible and infrared radiation
- Solar energy reaching the top of Earth's atmosphere, averaged globally, is approximately 340 watts per square meter (W/m^2)

EARTH'S ENERGY BUDGET

Energy Out

- **Reflection:** About 30% of the incoming solar energy is reflected into space by clouds, aerosols, and Earth's surface
- **Absorption & Emission:** The remaining 70% is absorbed by the atmosphere, oceans, and land
- This absorbed energy is then re-emitted as longwave (infrared) thermal radiation
- **Atmospheric Emission:** The atmosphere itself also emits radiation, with some of it flowing back to Earth which helps to warm the planet, and the rest going out to space

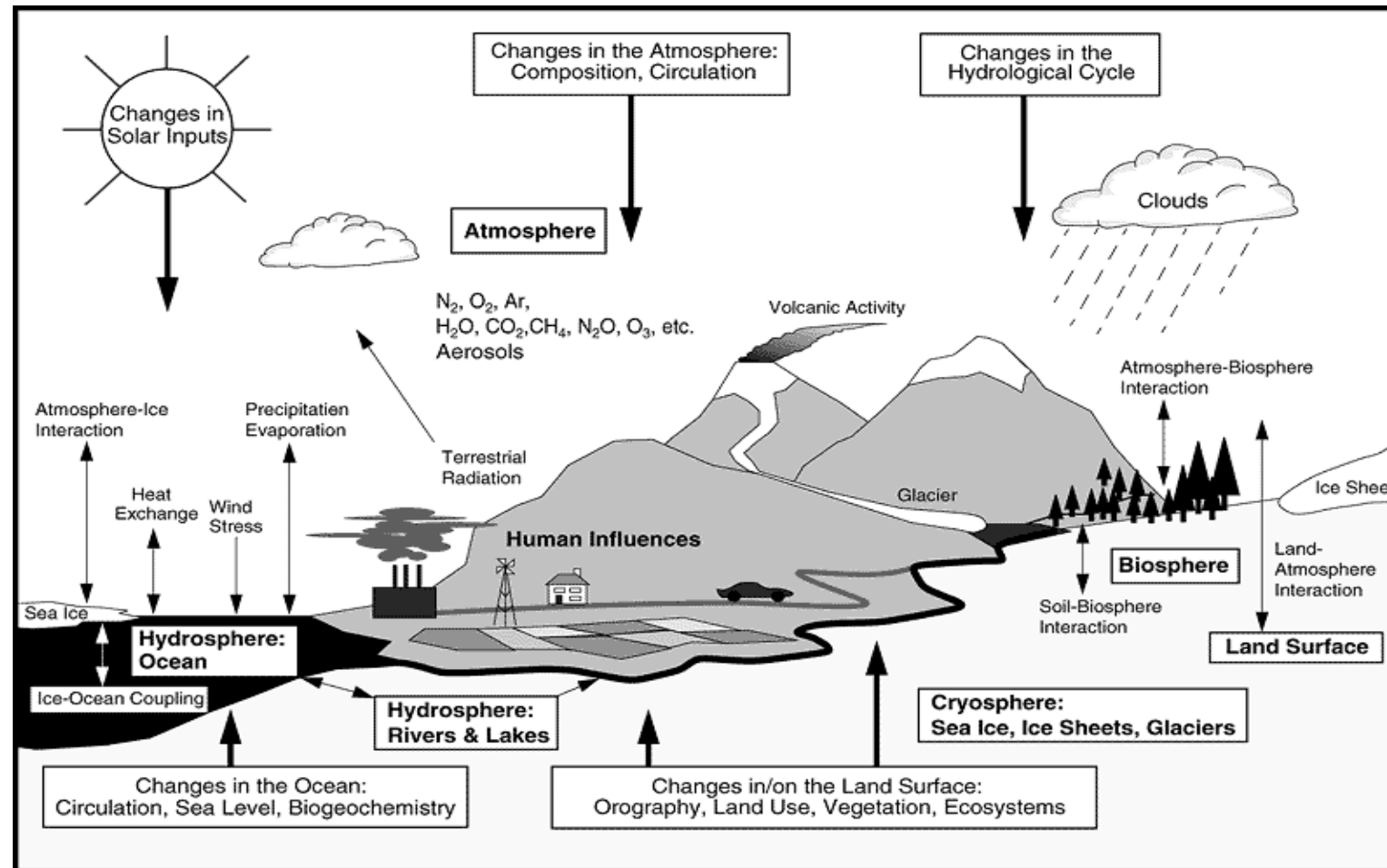




- Some surfaces are more reflective than others, characterized by the surface's albedo
 - Albedo ~ amount of energy reflected by a surface, measured on a scale from zero to one (or as a percent)
- Dark colored surfaces, like ocean and forests, reflect very little of the solar energy
- Light colored parts of the surface, like snow and ice, reflect almost all of the solar energy

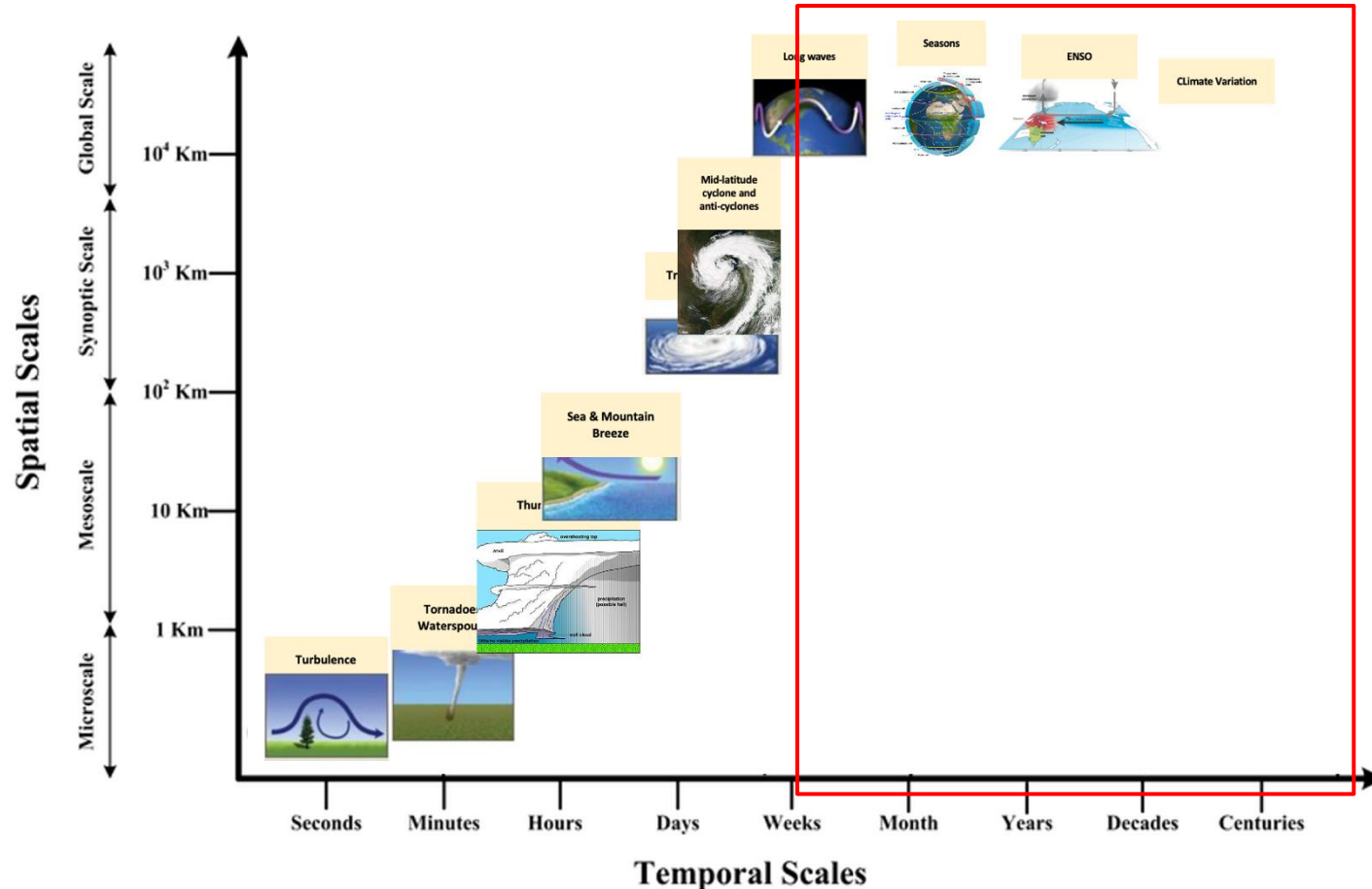
NATURAL CLIMATE VARIABILITY

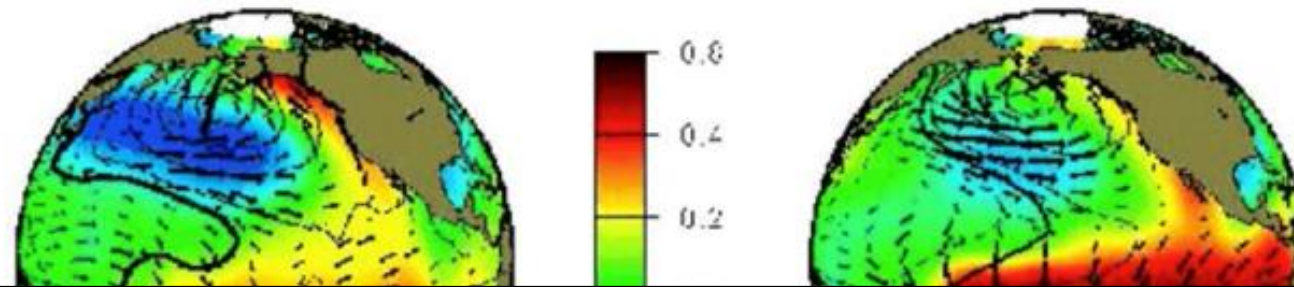
- The climate system evolves in time under the influence of its own internal dynamics
 - Without human-induced climate change ~ internal natural climate variability



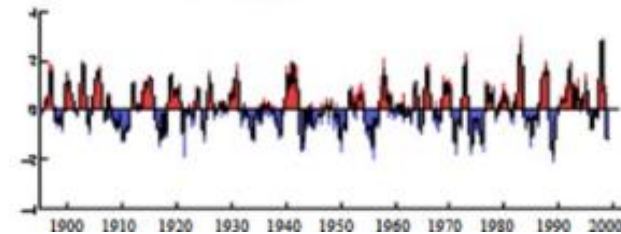
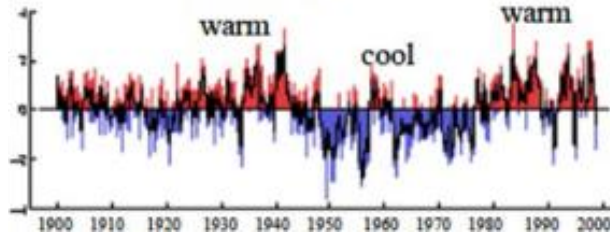
NATURAL CLIMATE VARIABILITY

- Natural fluctuations in the mean state and other statistics (standard deviations, the occurrence of extremes etc.) of the climate on all temporal and spatial scales beyond individual weather events
- Described as "**modes**" or "**oscillations**" that influence regional and global weather patterns
- Can occur on a variety of time scales: seasonal, year-to-year, decadal and beyond



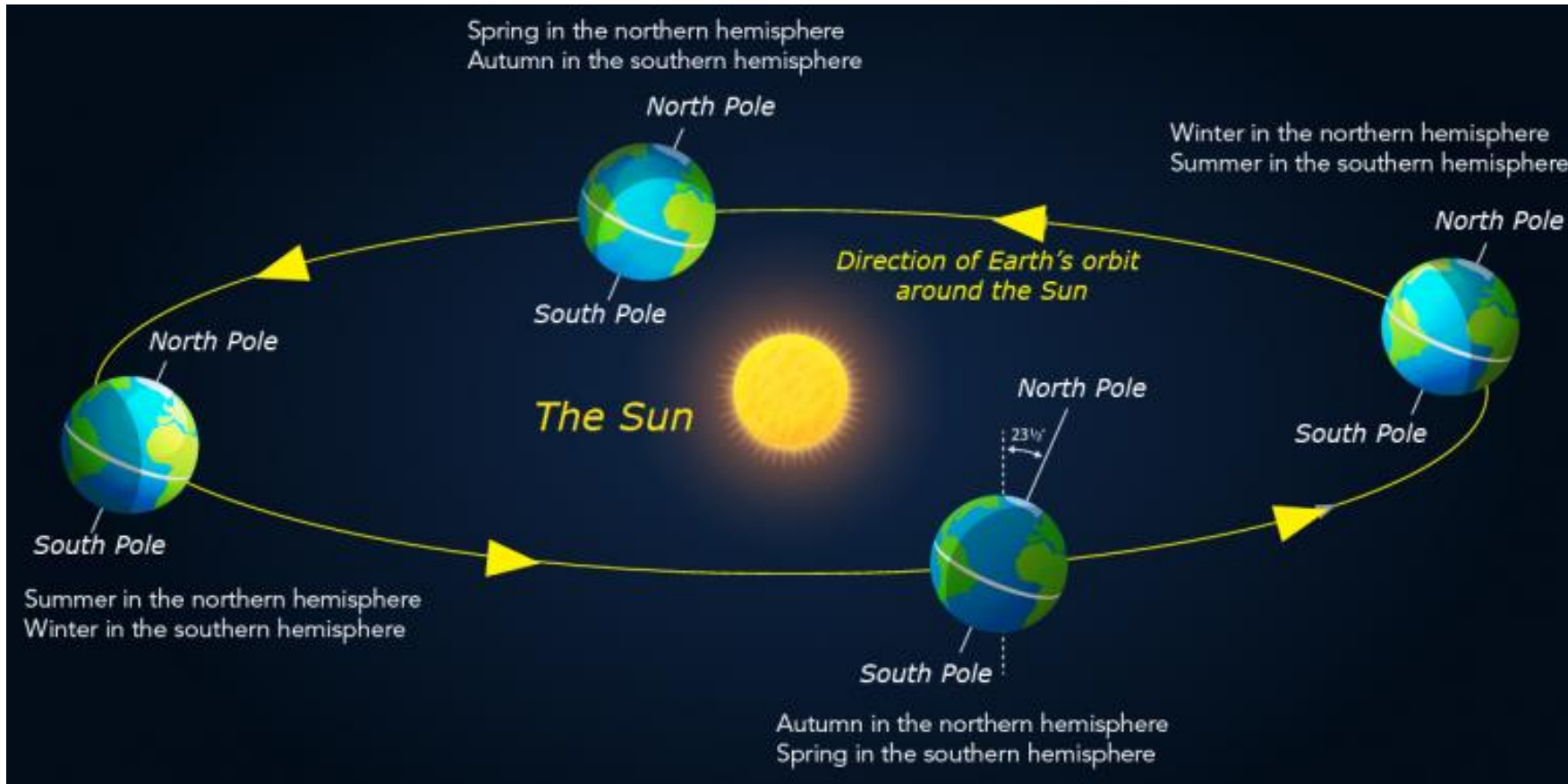


OSCILLATIONS AS DRIVERS OF NATURAL (INTERNAL) CLIMATE VARIABILITY



SEASONAL CYCLES

SHORTER CYCLES



- Recurring patterns of weather changes throughout the year
- Driven by the Earth's orbit around the sun and its axial tilt
- These cycles cause variations in temperature, precipitation, and other weather elements
 - lead to distinct seasons like spring, summer, autumn, and winter

SEASONAL CYCLES

SHORTER CYCLES

Monsoons

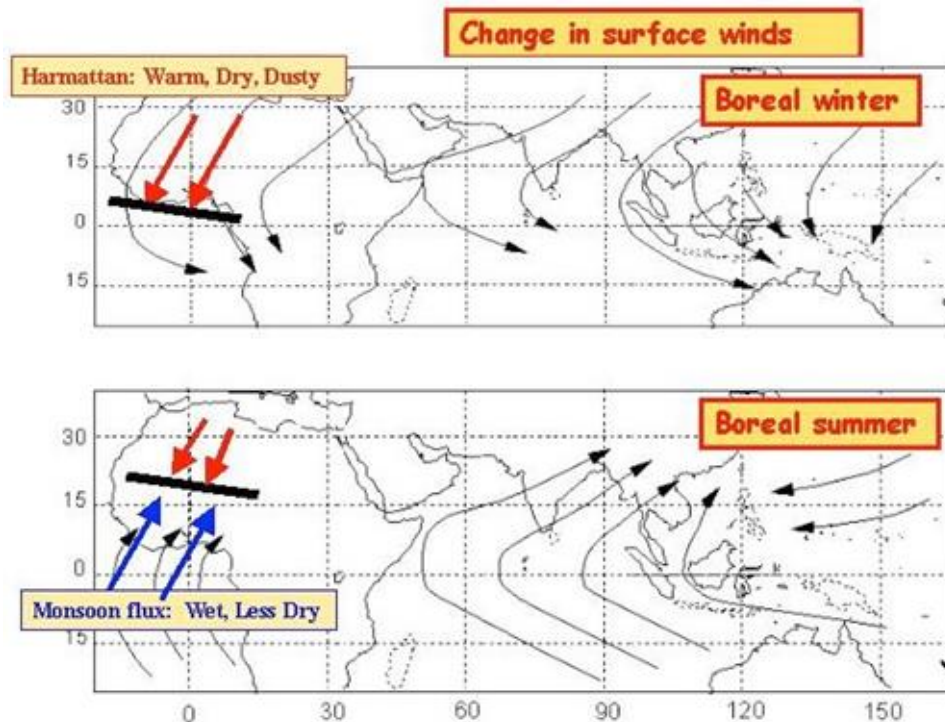
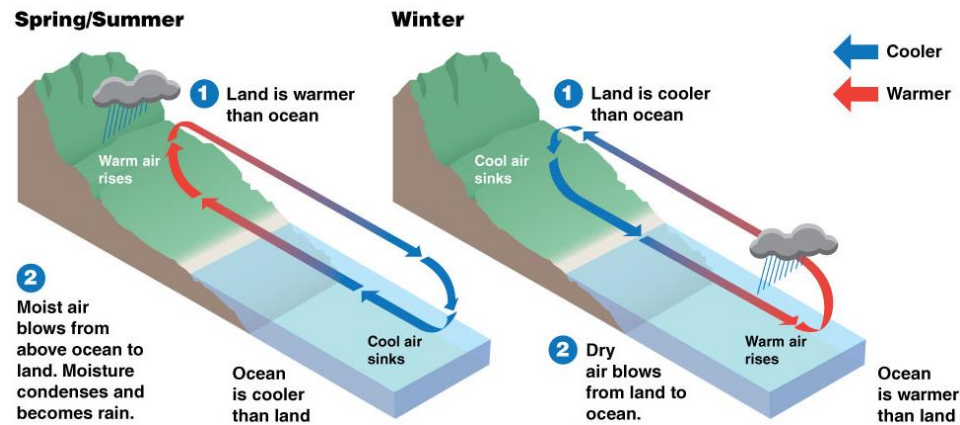
- Seasonal reversals of the wind direction that cause distinct wet and dry periods
- Definition includes annual weather cycle within the tropical and subtropical continents of Asia, Australia and Africa and the adjacent seas and oceans

Summer monsoon:

- In summer, the land heats up faster than oceans, creating low pressure over land and high pressure over the ocean
- This causes winds to blow from the ocean towards the land, bringing warm, moist air and heavy rainfall

Winter monsoon:

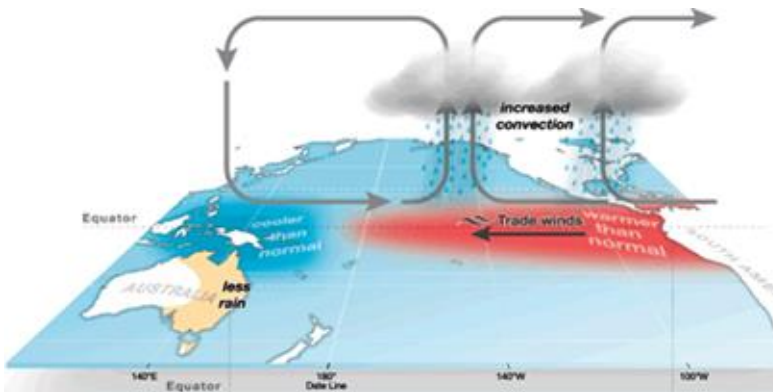
- In winter, the land cools down faster than the oceans, leading to high pressure over land and low pressure over the ocean
- This causes winds to blow from the land towards the sea, resulting in dry conditions



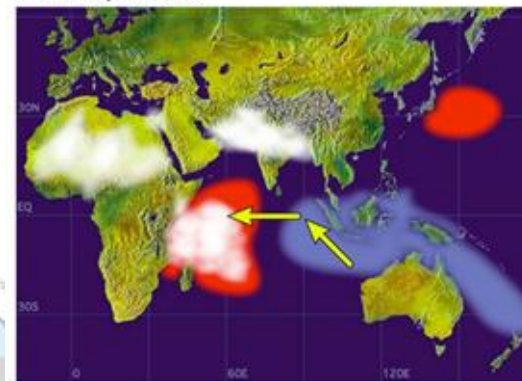
INTER ANNUAL VARIABILITY

SHORTER CYCLES

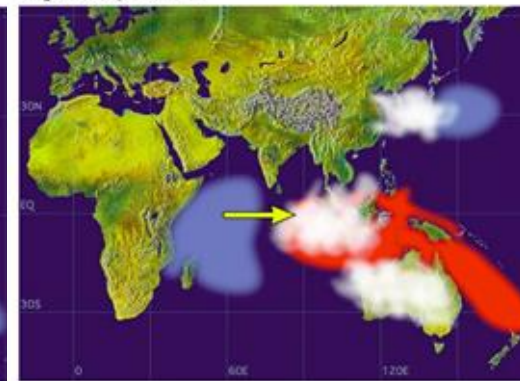
- Modify the seasons
- Longer time period (3 – 10+ years) variability often linked to slower changing ocean oscillations
 - El-Niño Southern Oscillation (ENSO)
 - Indian Ocean Dipole (IOD)
 - Southern Annular Mode (SAM)
- Some processes/oscillations are decadal and longer.....



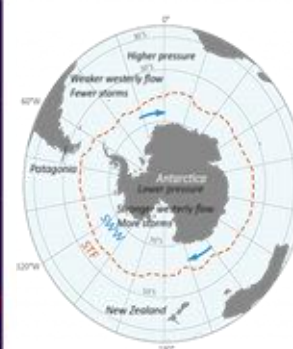
Positive Dipole Mode



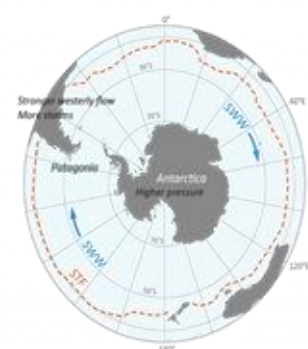
Negative Dipole Mode



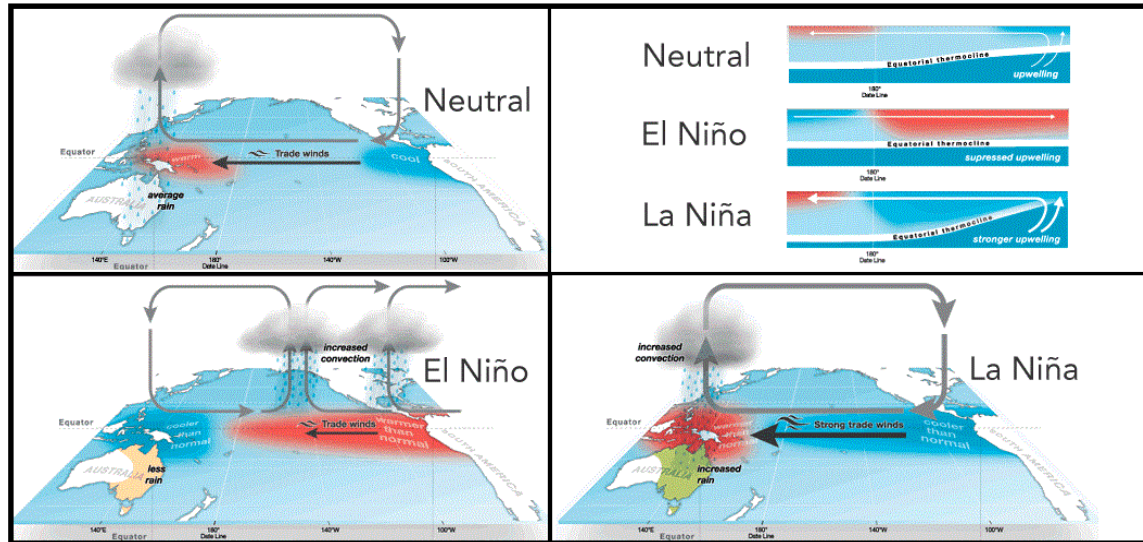
POSITIVE SOUTHERN ANNULAR MODE



NEGATIVE SOUTHERN ANNULAR MODE



El Nino Southern Oscillation (ENSO)

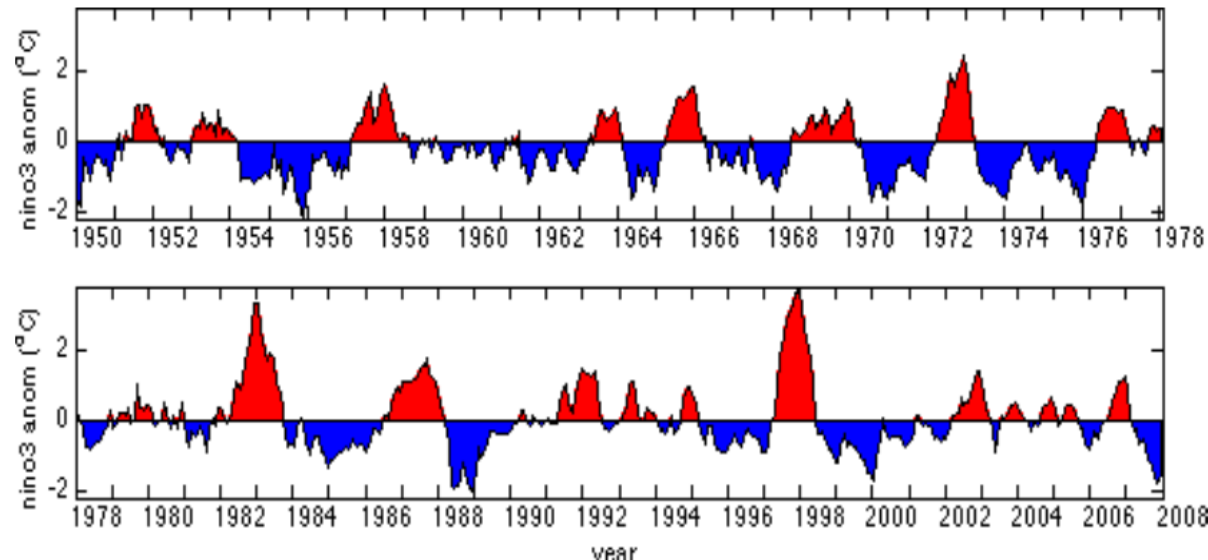


What is it?

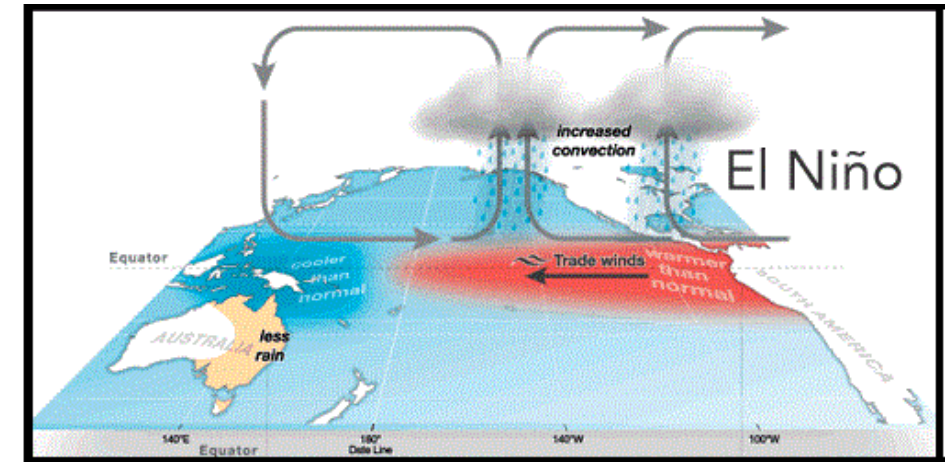
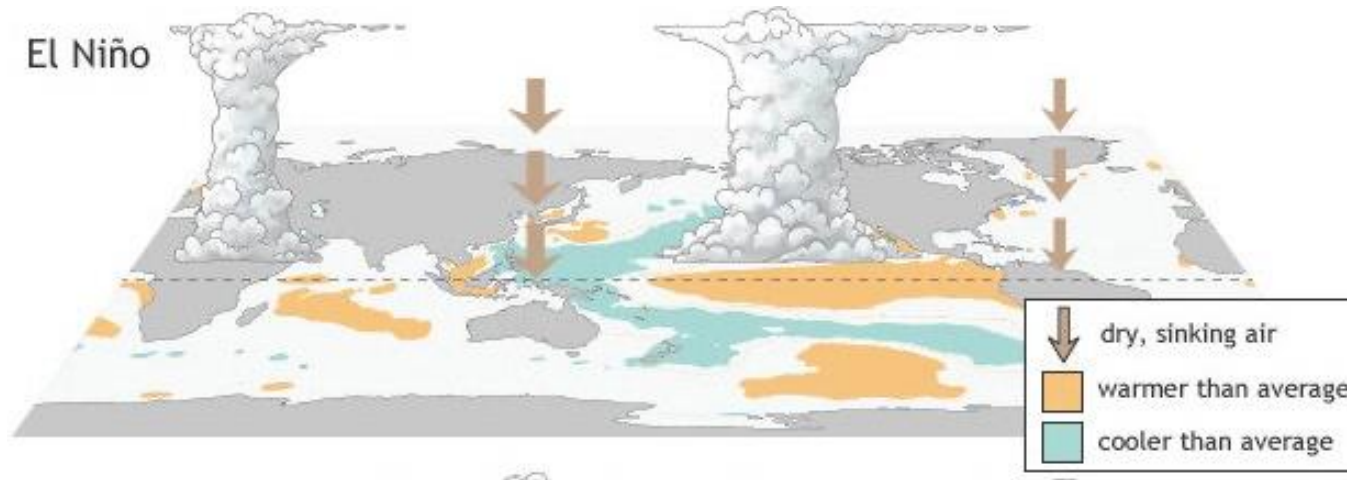
- Naturally occurring climate pattern involving fluctuating ocean temperatures in the tropical Pacific and changes in the atmosphere

What is the time scale?

- Inter-Annual Variation: 3-6 year cycle

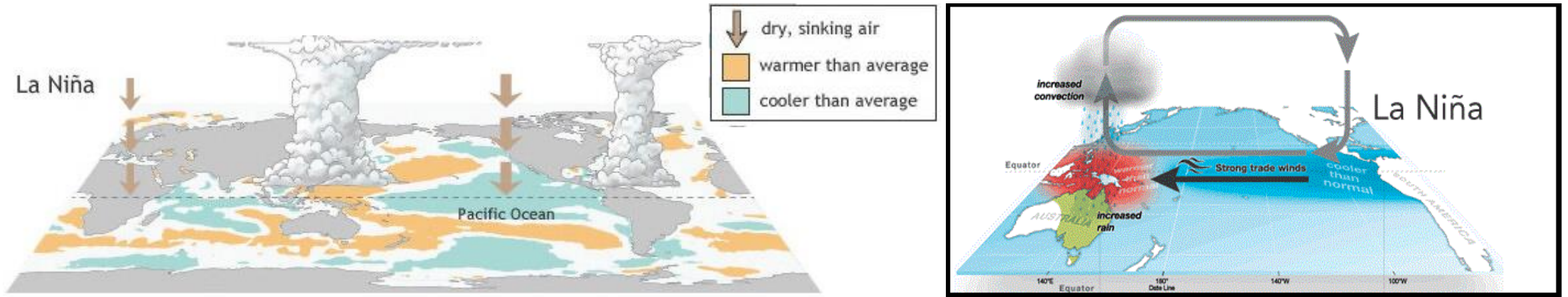


El Nino Southern Oscillation (ENSO)



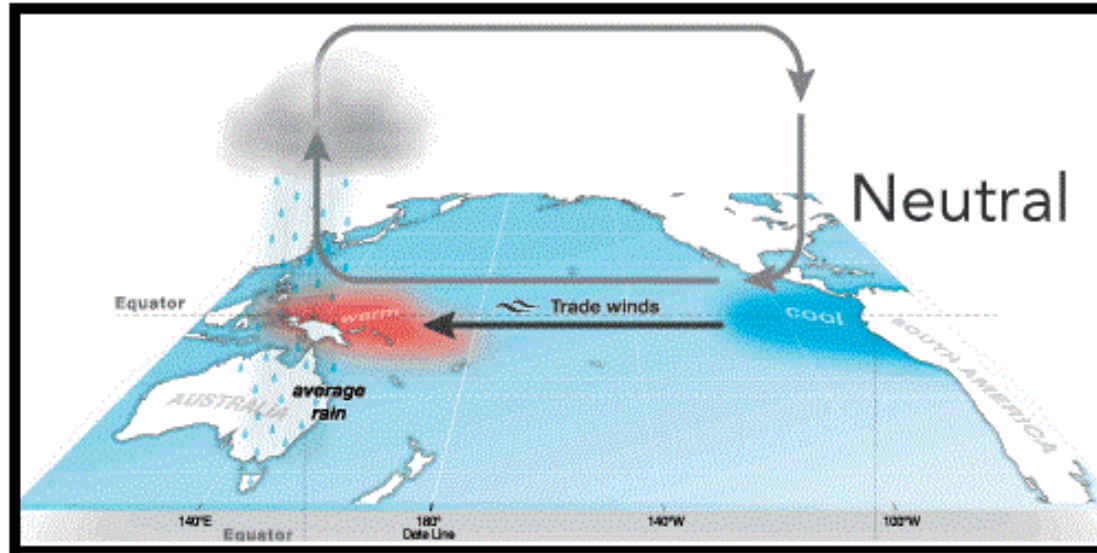
- Warmer-than-average sea surface temperatures (SST) in the central and eastern tropical Pacific Ocean
 - Due to a weakening of the westward blowing trade winds (easterly trade winds)
- Trade winds: the low-level surface winds, which normally blow from east to west along the equator
- Warm water from western Pacific surges back towards the east
- Often peaks during the Northern Hemisphere winter
- Upwelling of cooler water along eastern equatorial Pacific coastline slows or stops

El Nino Southern Oscillation (ENSO)



- Characterized by cooling of the ocean surface, or below-average sea surface temperatures (SST), in the central and eastern tropical Pacific Ocean
 - Due to stronger, than normal, easterly trade winds that push warm water to the west
- Increased upwelling of cool waters along eastern equatorial Pacific and more cold water being pushed further across the Pacific

El Nino Southern Oscillation (ENSO)



- Neither El Niño nor La Niña conditions are present
- Typical trade winds blow from east to west across the tropical Pacific, pushing warm surface water towards the western Pacific and allowing cold, nutrient-rich water to upwell in the eastern Pacific

INTER ANNUAL VARIABILITY

SHORTER CYCLES

El Nino Southern Oscillation (ENSO)

Impacts

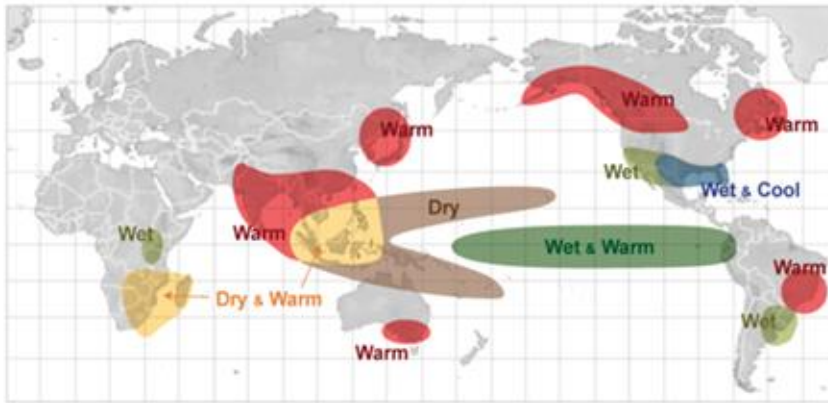


Figure 4: El Niño climate impacts December – February

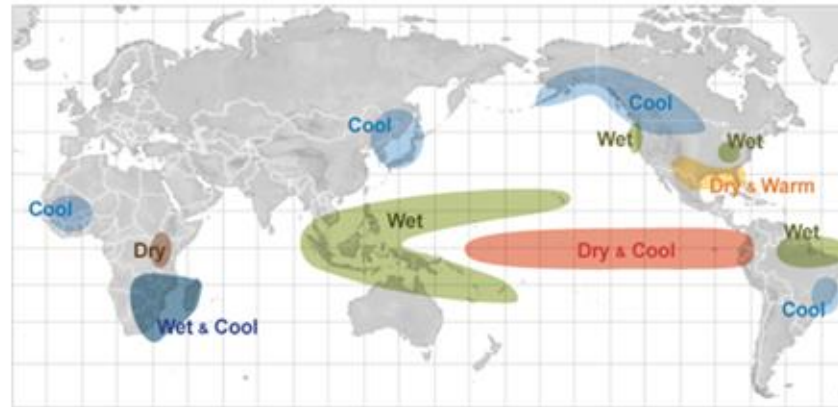


Figure 6: La Niña climate impacts December – February

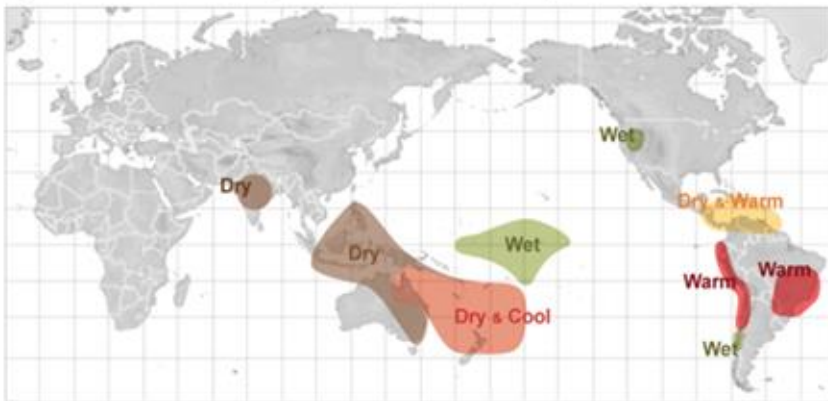


Figure 5: El Niño climate impacts June – August

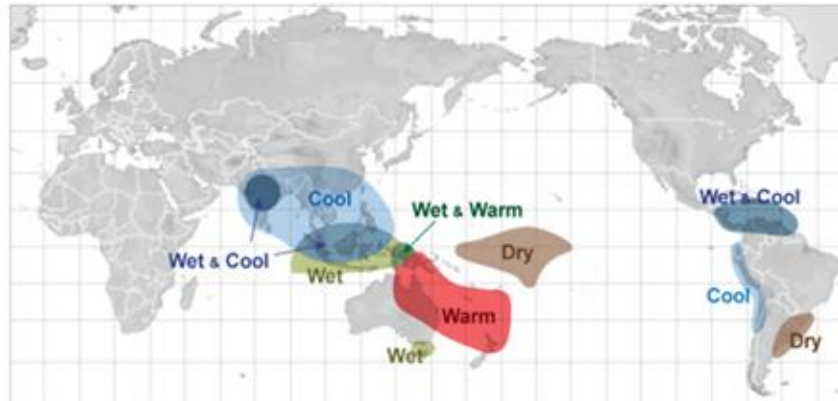


Figure 7: La Niña climate impacts June – August

Drier conditions

- Droughts
- Increased risk of wildfires
- Disrupted marine ecosystems

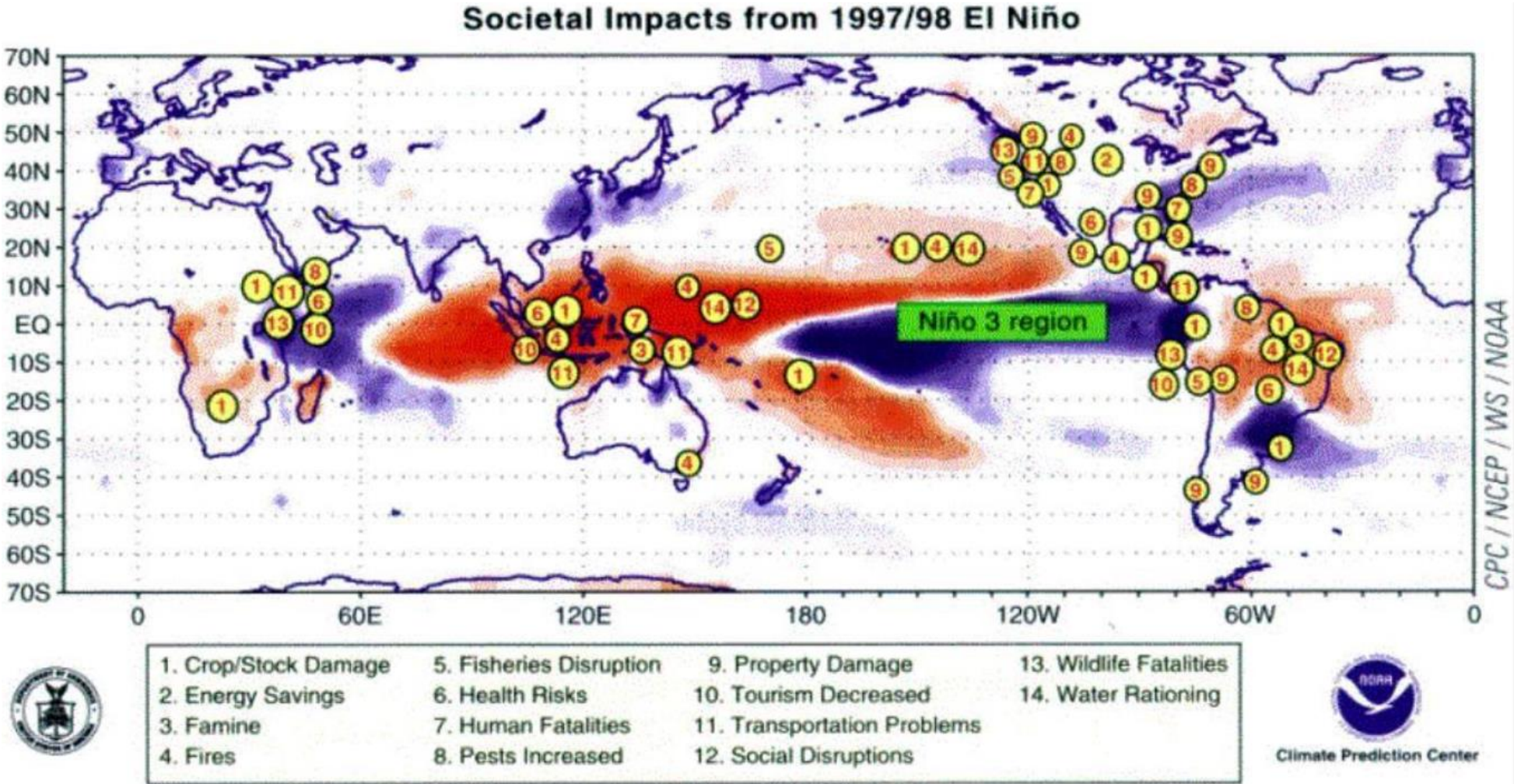
Wet conditions

- Increased rainfall and flooding

Health impacts (in dry or wet)

- Alter transmission of diseases
- Increased outbreaks of waterborne and vector-borne illnesses

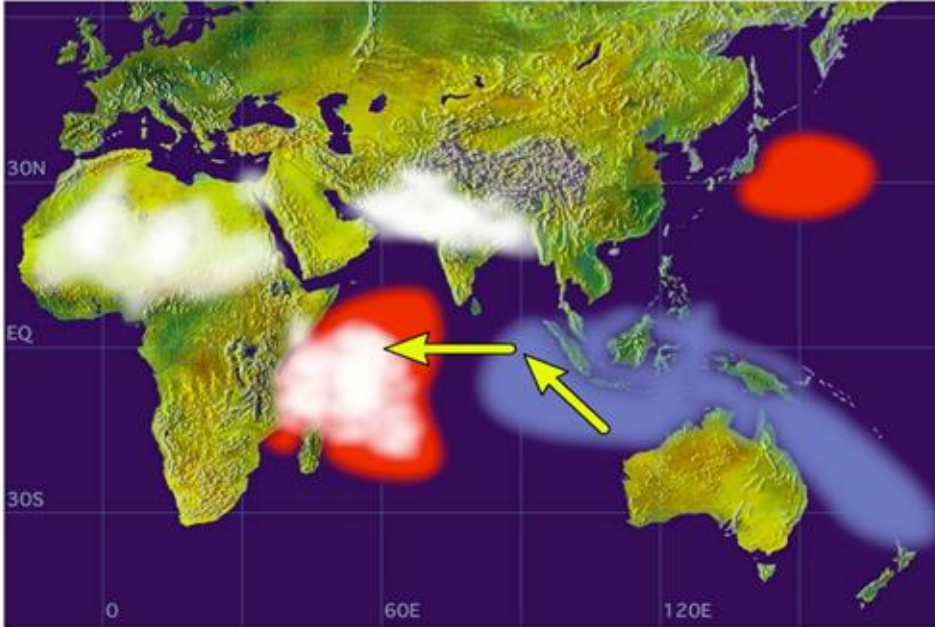
El Nino Southern Oscillation (ENSO)



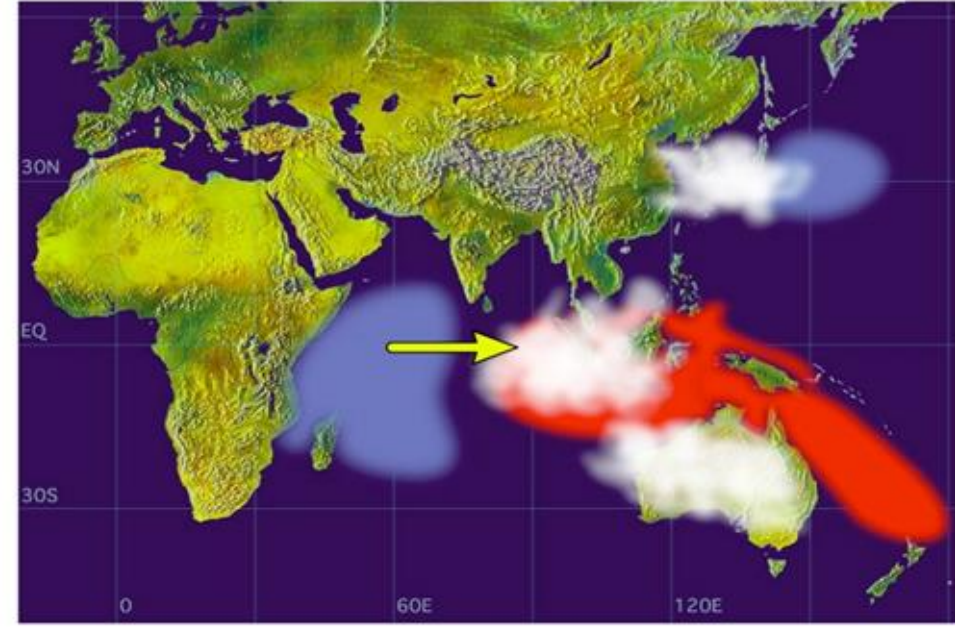
Red: rainfall below normal, blue: rainfall above normal. Compare the observed rainfall pattern with that predicted by the coupled ocean-atmosphere model; see page 23.

Indian Ocean Dipole (IOD)

Positive Dipole Mode



Negative Dipole Mode



What is it?

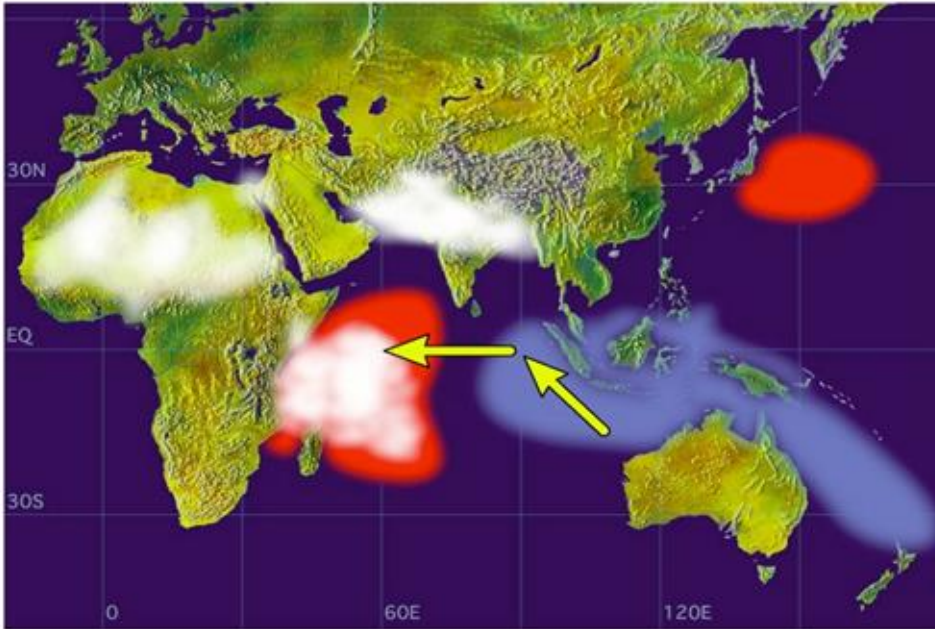
- A climate pattern in the Indian Ocean characterized by fluctuating SSTs between the western and eastern tropical regions

What is the time scale?

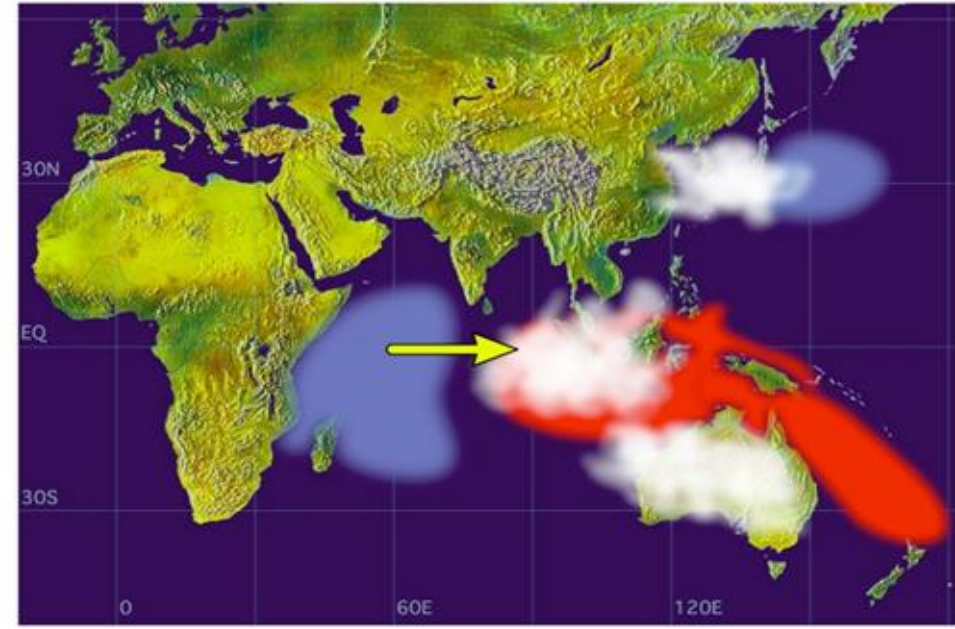
- Inter-Annual Variation: 4 – 6 year cycle

Indian Ocean Dipole (IOD)

Positive Dipole Mode

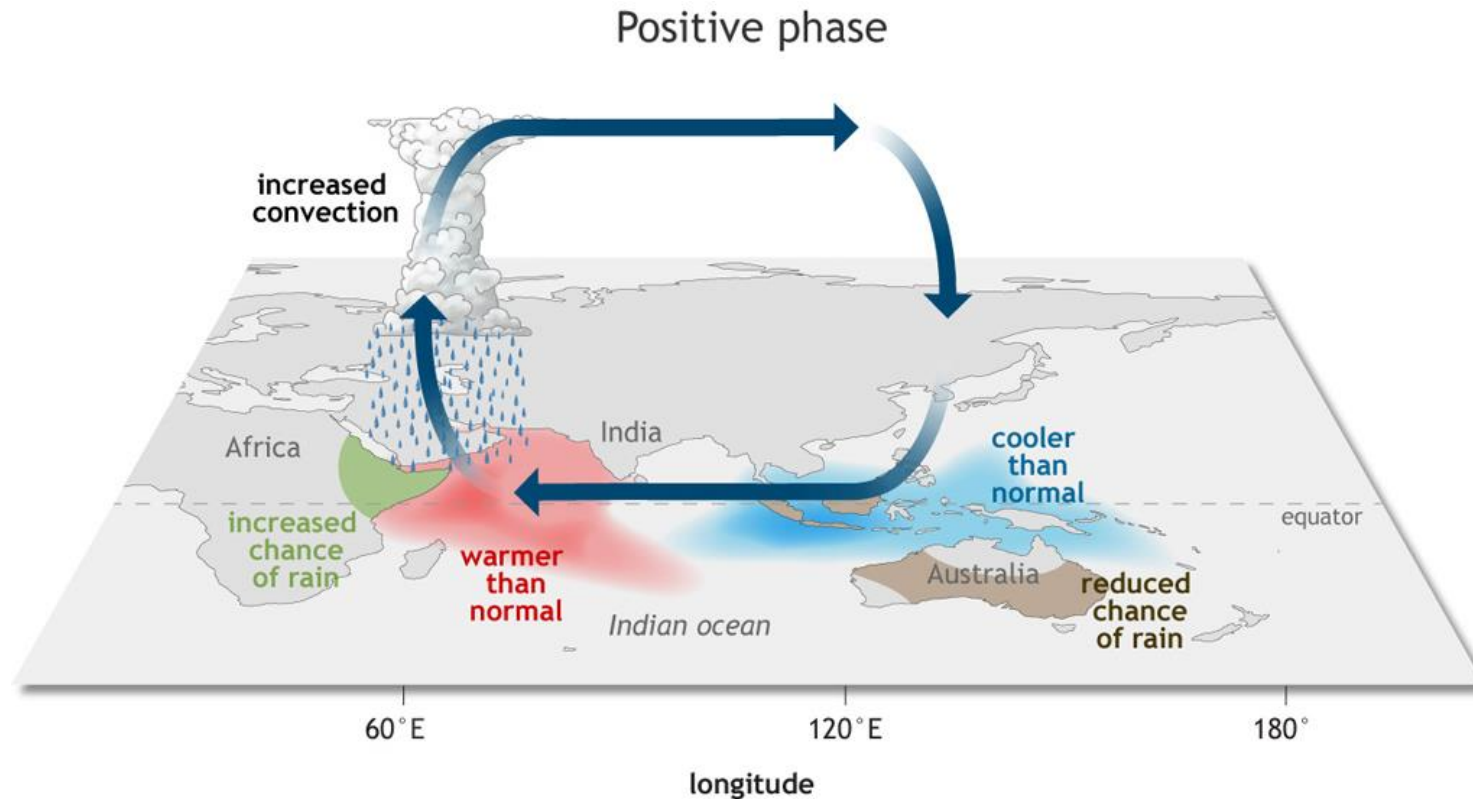


Negative Dipole Mode



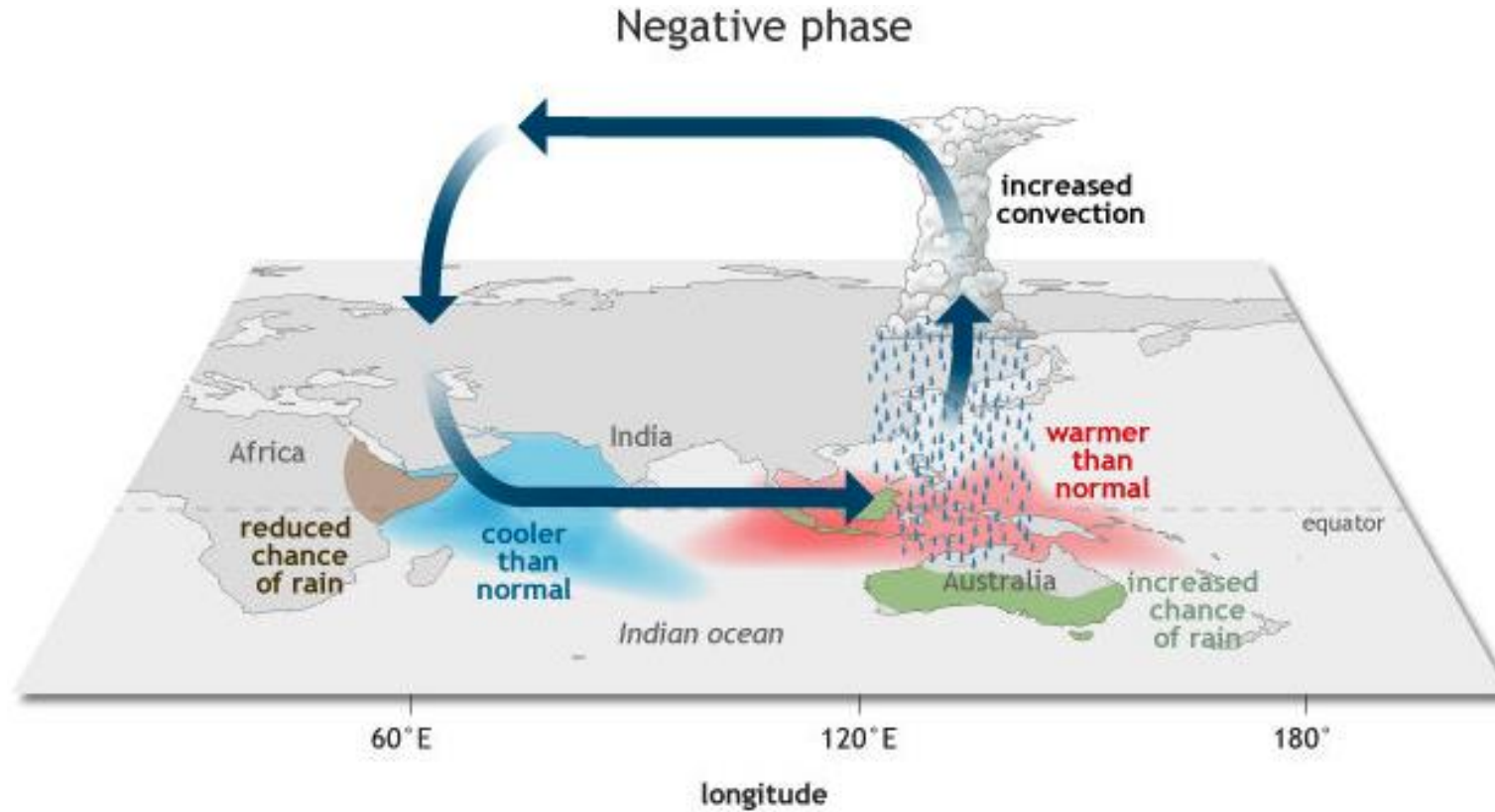
- Involves an oscillation of SSTs, where the western Indian Ocean becomes warmer (or colder) than the eastern part, and vice versa
- IOD events typically start around May or June, peak between August and October, and then weaken as the monsoon arrives in the Southern Hemisphere

Indian Ocean Dipole (IOD)



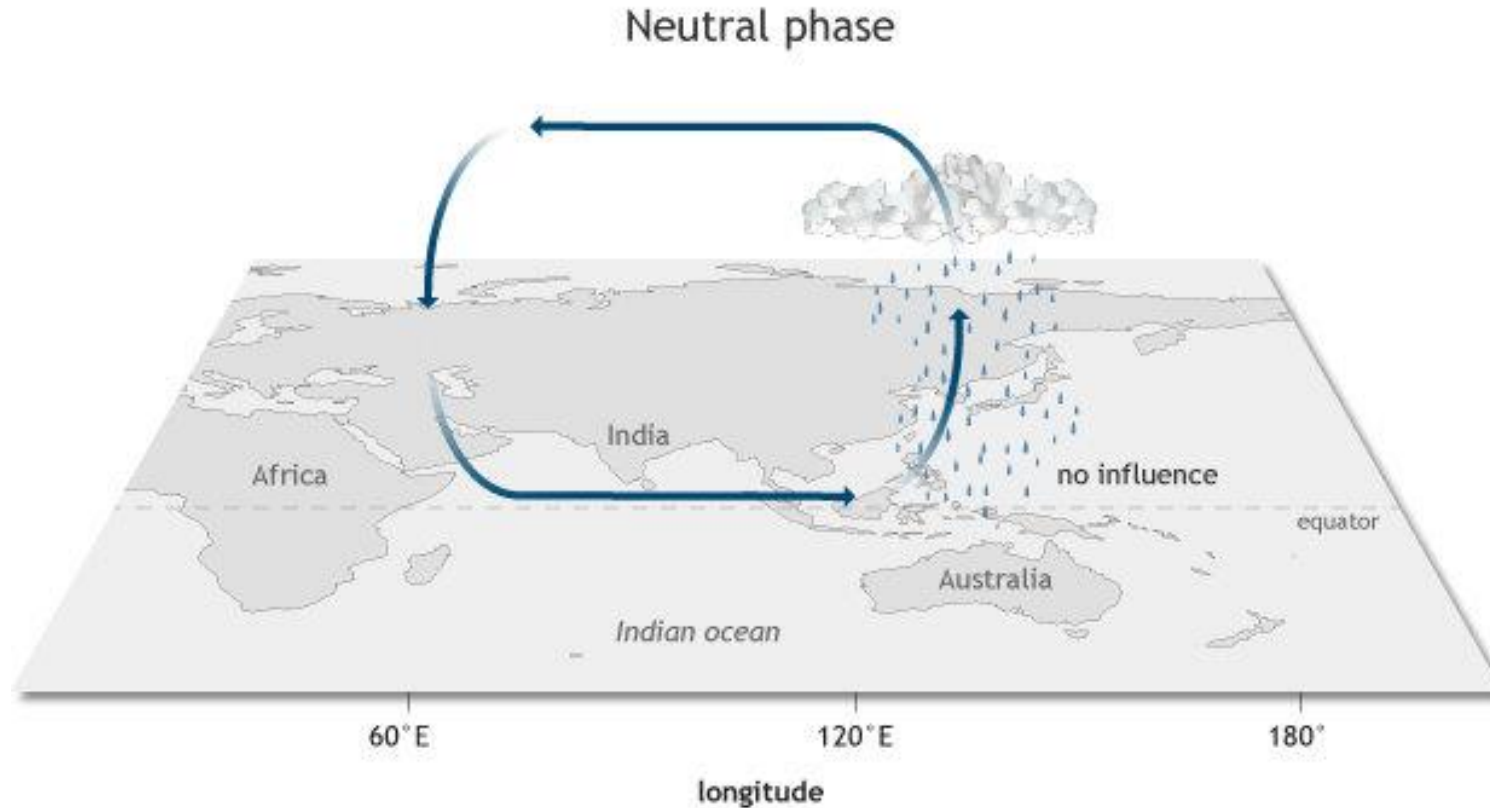
- Warmer-than-average waters in the western Indian Ocean and cooler-than-average waters in the east
- Indonesia and Australia tend to be drier than normal, which increases the chances of bushfires
- Eastern Africa tends to be wetter than normal, increasing the likelihood of floods

Indian Ocean Dipole (IOD)



- Cooler-than-average waters in the west and warmer-than-average waters in the east
- Indonesia and Australia tend to be wetter than normal, while eastern Africa tends to be drier than normal

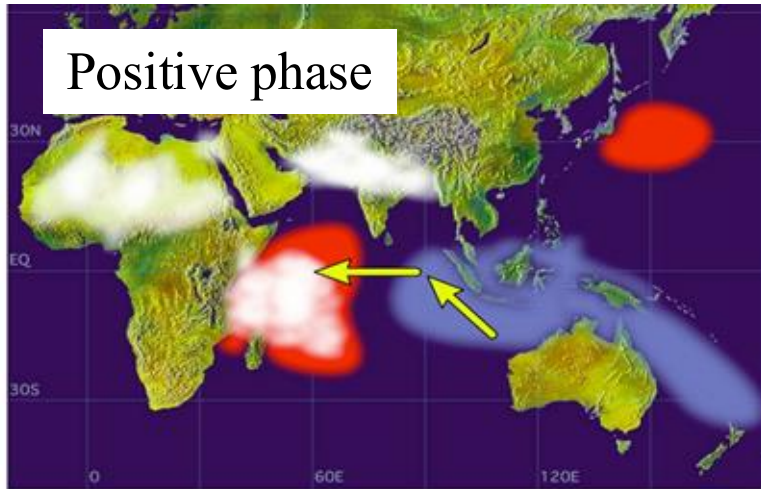
Indian Ocean Dipole (IOD)



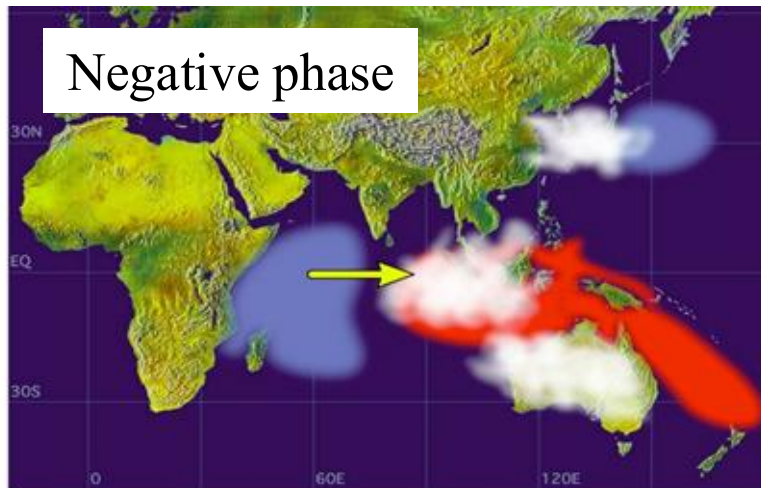
- The neutral phase of the Indian Ocean Dipole shows neither positive nor negative features

Indian Ocean Dipole (IOD)

Impacts

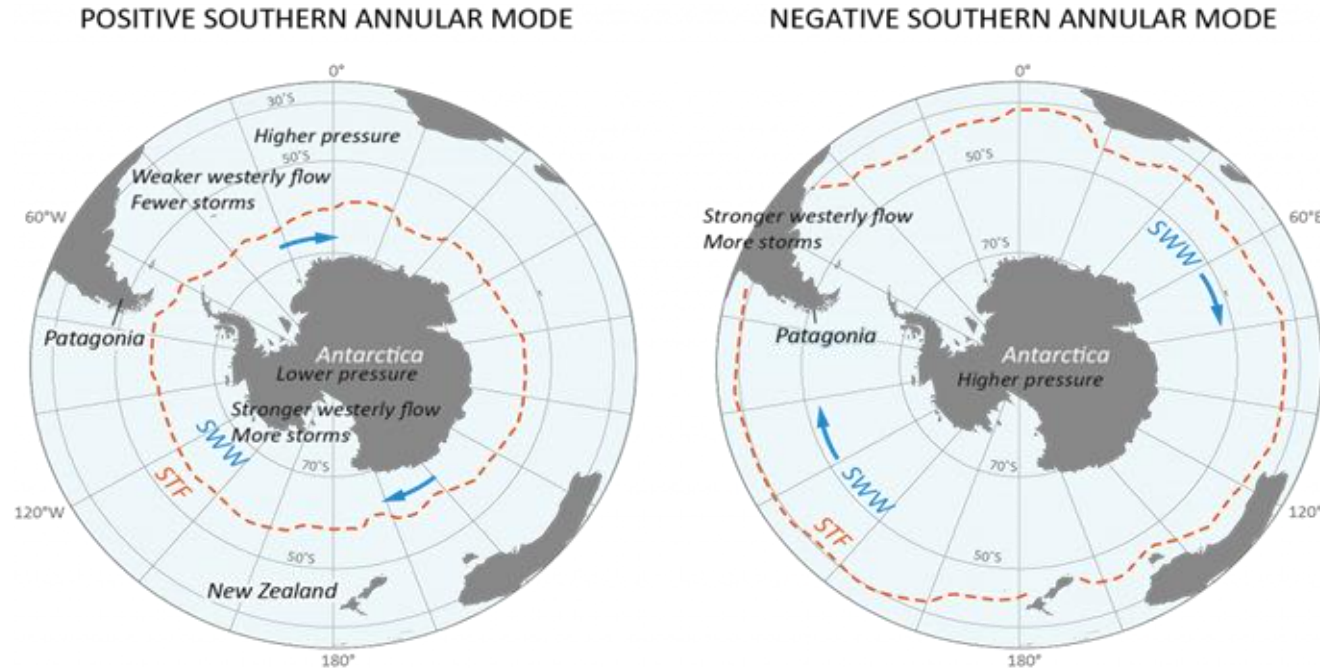


- Warmer-than-average SSTs and greater precipitation in the western Indian Ocean region
- Corresponding cooling of waters in the eastern Indian Ocean
- Cause droughts in adjacent land areas of Indonesia and Australia and heavy rainfall over east Africa



- Opposite conditions with warmer water and greater precipitation in the eastern Indian Ocean
- Cooler and drier conditions in affected African regions
- Affects the strength of monsoons over the Indian subcontinent

Southern Annular Mode (SAM)



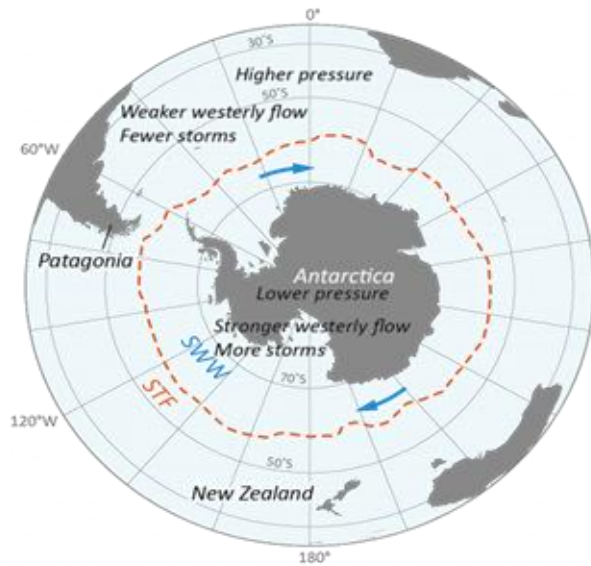
What is it?

- The north-south shifts in the belt of strong westerly winds that encircle Antarctica
- The difference in sea-level pressure between mid-latitudes ($\sim 40^\circ\text{S}$) and high latitudes ($\sim 65^\circ\text{S}$)
- Changes in air pressure reflect the strength and position of the belt of westerly winds that circle Antarctica

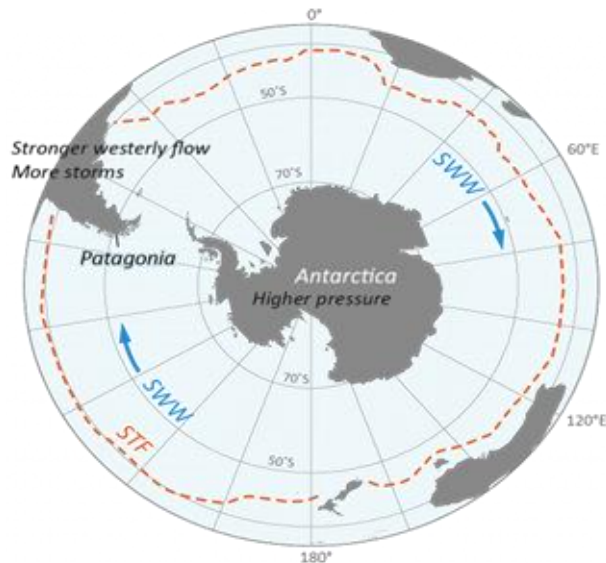
What is the time scale? 10 – 30 days

Southern Annular Mode (SAM)

POSITIVE SOUTHERN ANNULAR MODE



NEGATIVE SOUTHERN ANNULAR MODE



- Lower anomalous air pressure over Antarctica
 - Higher anomalous air pressure over the mid-latitudes
 - The belt of strong westerly winds strengthens and contracts towards Antarctica
 - Weakens at the northern boundary in the mid-latitudes (40-50°S)
-
- The belt of strong Southern Westerly Winds expands northwards towards the equator, bringing cold, wet weather to Patagonia
 - The winds are weaker in this phase

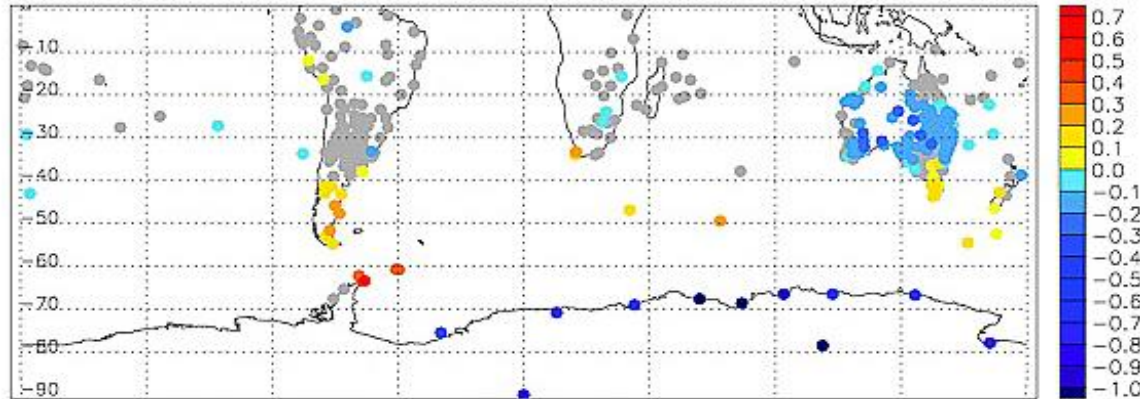
Southern Annular Mode (SAM)

Impacts

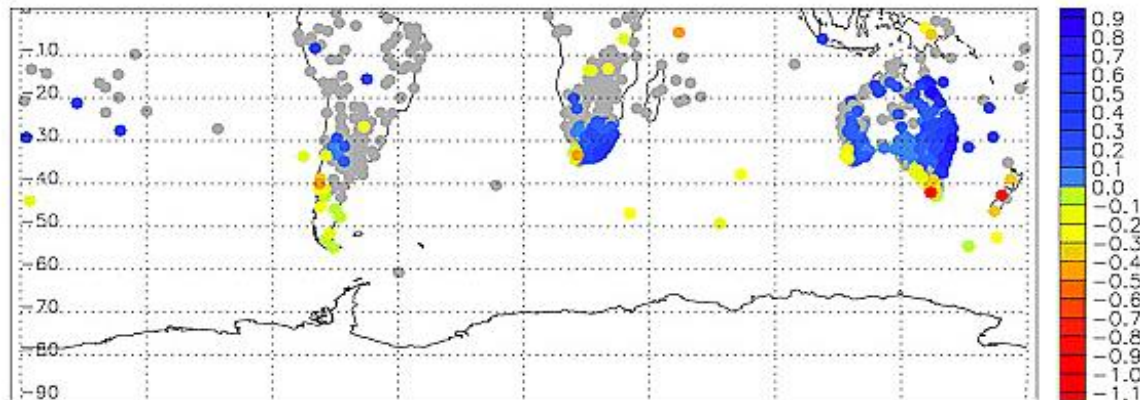
A positive phase of SAM leads to:

- Increased rainfall over parts of Southern Africa and Australia
- Decreased rainfall over Southern parts of Australia, South-West of South Africa, New Zealand and Southern parts of South America
- Warming over Southern Parts of South America, southern parts of Australia and New Zealand, over South West South Africa and the Antarctic Peninsula

(b)



(c)

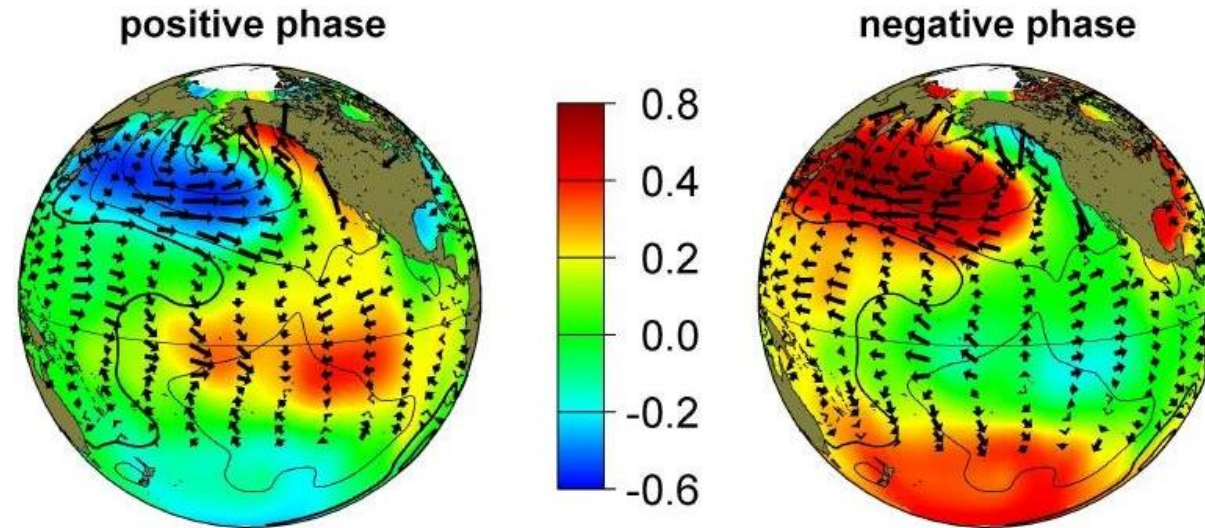


“Low Frequency Variability”

- These cycles influence the shorter time scale variability
- More difficult to observe and characterize as a result of poorer observational records the further we go back in time
- How they influence the shorter time scale cycles is often not well understood

“Low Frequency Variability”

Pacific Decadal Oscillation (PDO)



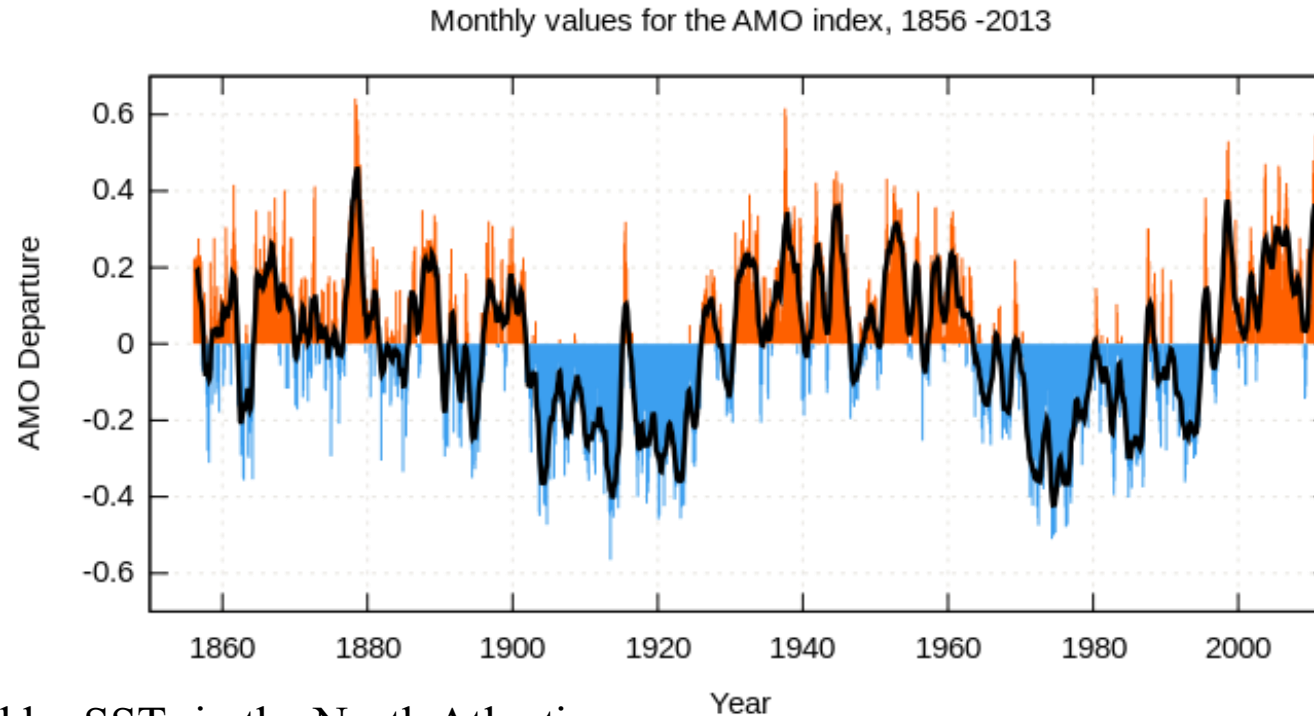
- Defined by SST anomalies in the higher latitudes of the Pacific
- While PDO also has a warm phase and cool phase, the timescale on which PDO fluctuates is not as defined as ENSO
- It can persist in either phase for anywhere from 10-40 years
- During the warm, or positive, phase, the northwest Pacific is relatively cooler and the eastern Pacific warmer
- The inverse pattern appears during the cool, or negative, phase (causes of these patterns are not clearly understood)
- Modulates ENSO or possibly ENSO modulates it

DECADAL AND INTER-DECADAL VARIABILITY

SHORTER CYCLES

“Low Frequency Variability”

Atlantic Multidecadal Oscillation (AMO)

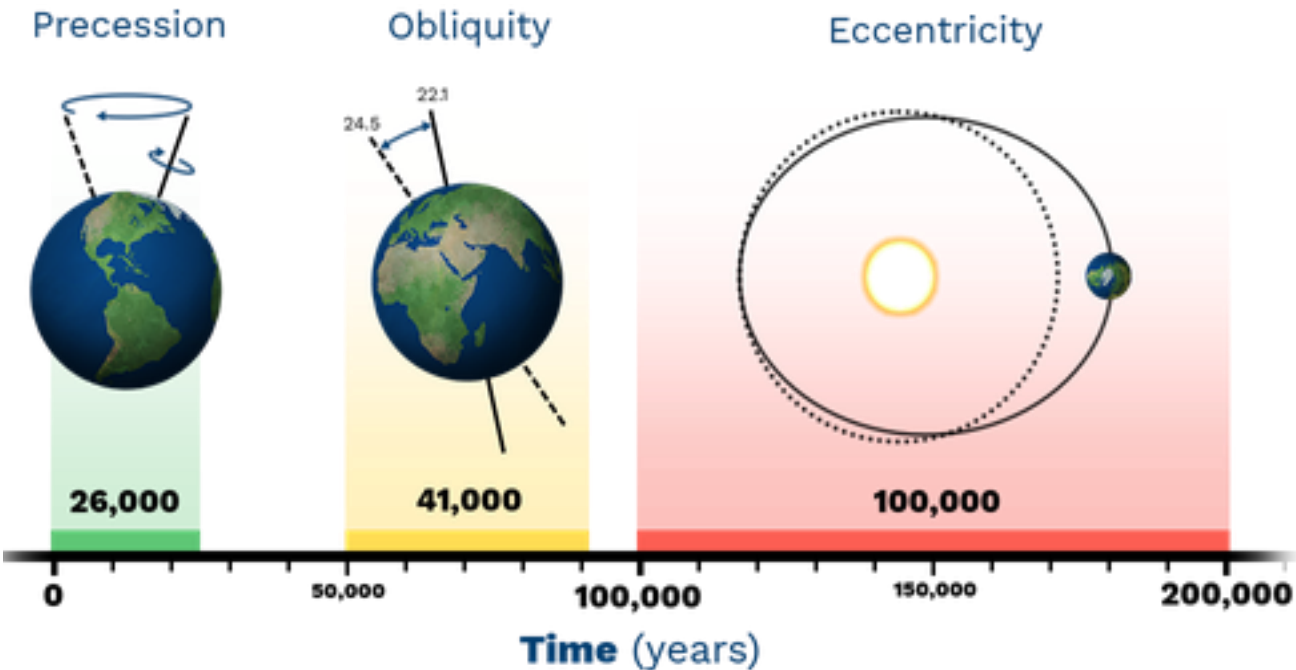


- Defined by SSTs in the North Atlantic
- It is a longer-term source of variability (70 year cycle)
- Only a few shifts between its warm and cool phases in the 20th century
- Linked to temperature and rainfall patterns in North America and Europe, as well as Brazil, the Sahel and India
- Influences hurricane activity in the Atlantic

MILANKOVITCH CYCLES

LONGER CYCLES

- Very long cycles: Thousands of years
- Can increase seasonal amplitudes (intensity of seasonal variation)
- Can favor ice ages



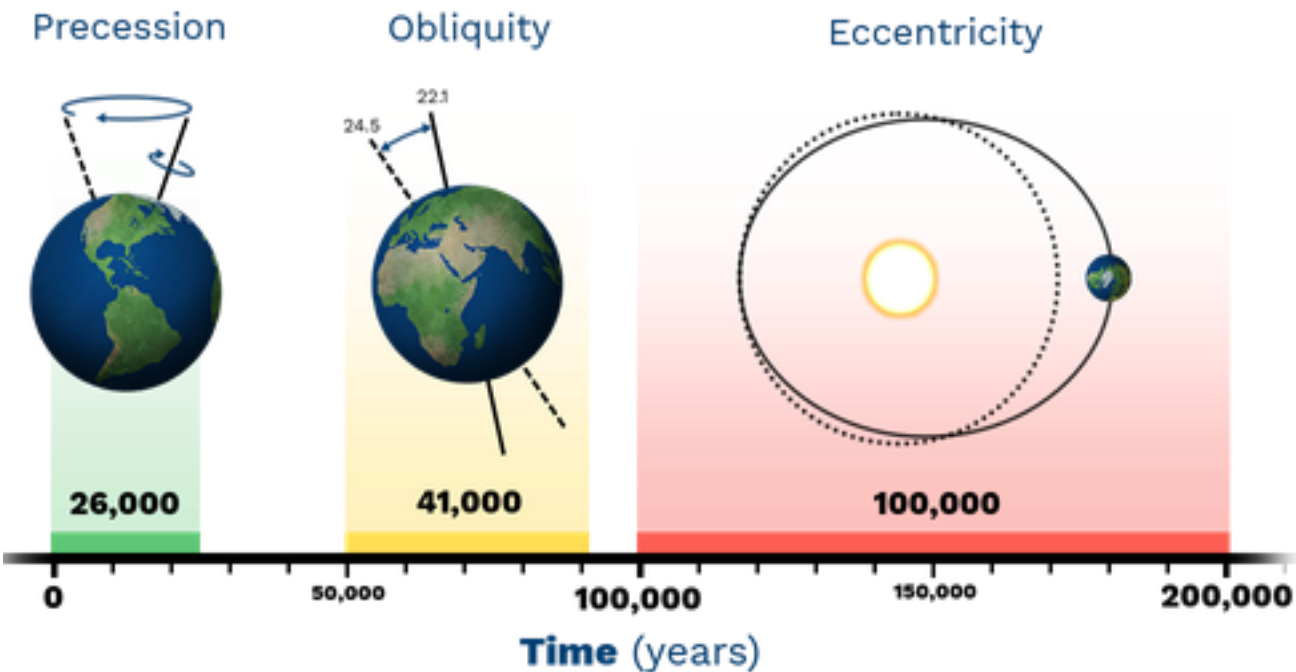
Eccentricity

- The Earth's orbit around the Sun is not a perfect circle, but rather an ellipse, and its shape changes over time
- This cycle takes about 100,000 years to complete
- When the orbit is more elliptical, Earth experiences greater variations in solar radiation between its closest and furthest points from the Sun

MILANKOVITCH CYCLES

LONGER CYCLES

- Very long cycles: Thousands of years
- Can increase seasonal amplitudes (intensity of seasonal variation)
- Can favor ice ages



Obliquity

- The Earth's axial tilt, currently at 23.5 degrees, varies over time, ranging from about 22.1 to 24.5 degrees
- This cycle takes roughly 41,000 years to complete
- Increased tilt leads to more extreme seasons, while a smaller tilt results in milder seasons

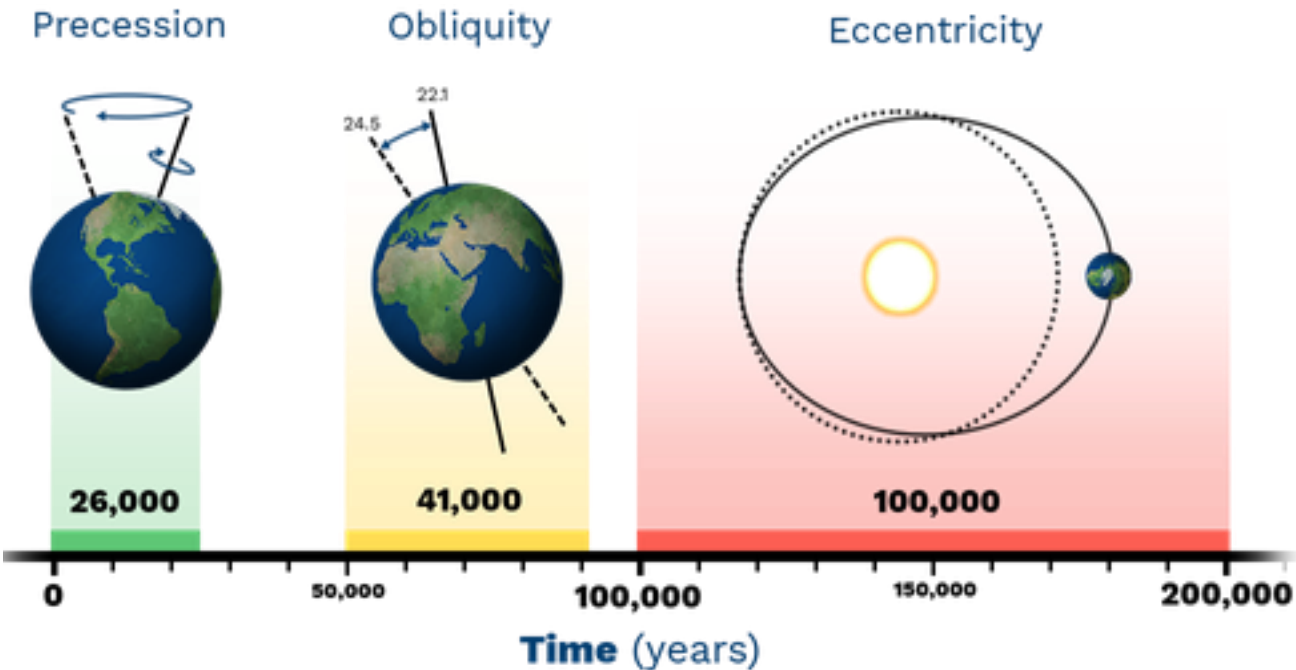
MILANKOVITCH CYCLES

LONGER CYCLES

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Precession

- Refers to the slow wobble of Earth's axis, similar to a spinning top
- This wobble is influenced by the gravitational pull of the Sun and Moon
- Cycle has a period of about 26,000 years
- This wobble affects the timing of the seasons in relation to Earth's orbit



NATURAL CLIMATE VARIABILITY

- There is extensive natural variability in the earth-atmosphere-ocean system
- These occur on many time scales, and they modulate each other
- We do not understand the mechanisms of many of the natural oscillations

Challenges:

- Are there cycles we have not discovered yet?
- How do we filter the effects of natural cycles in our weather from those effects caused by greenhouse gas emissions?
- How are these natural cycles affected by increasing concentrations of greenhouse gasses in the atmosphere?

NATURAL CLIMATE VARIABILITY

Name	What is it	Time Scale	Impacts
ENSO	Changes in wind and SSTs over tropical pacific	3-6 years	Has a strong influence on weather across the world. Over SA during Summer El Nino - warm & dry, La Nina - wet & cool
Indian Ocean Dipole (IOD)	Changes in wind and SSTs over Indian ocean	4-6 years	Positive Phase: Tends to cause droughts in adjacent land areas of Indonesia and Australia and heavy rainfall over east Africa. Negative Phase: Cooler and drier conditions in affected African regions.
Southern Annular Mode (SAM)	Changes in pressure over Antarctica and shifts in westerly winds	Weeks, sometimes years	Positive phase: Increased rainfall over parts of Southern Africa. Decreased rainfall and warming over South-west SA Negative phase:
Pacific Decadal Oscillation (PDO)		15-30 years	Modulates ENSO or possibly ENSO modulates it
Atlantic Multi-decadal Oscillation (AMO)		70 year	Obscures or exaggerates global temperature increases due to human-induced climate change
Milankovitch cycles		1000 of years	Can increase Seasonal Amplitudes. Can favor ice ages



THANK YOU